

**Writing Small Numbers in Scientific Notation***Question Bank - Additional 3.09*

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*Show your work. Circle your final answer.*

**For questions 1 through 4, write each number in scientific notation.**

1. 0.008

2. 0.00000026

3. 0.0475

4. 0.000000000091 seconds

**For questions 5 and 6, write each number in standard form.**

5.  $7 \times 10^{-5}$

6.  $2.6 \times 10^{-8}$

**Powers of ten.**

7. Write each value as a single power of 10, then as a fraction or decimal.

a.  $10^{-16}$ , as a fraction

b.  $\left(\frac{1}{10^{-5}}\right)^3$

c.  $10^{-9} \cdot 10^6$

8. Two measurements are  $7.2 \times 10^{-4}$  and 0.00000018.

a. Write  $7.2 \times 10^{-4}$  in standard form.

b. Write 0.00000018 in scientific notation.

c. Which of the two is greater? Explain how you know.

**9.** Complete each comparison.

**a.** How many times larger is 8.4 than 2.1?

**b.** How many times larger is  $10^{-3}$  than  $10^{-7}$  ?

**c.** How many times larger is  $8.4 \times 10^{-3}$  than  $2.1 \times 10^{-7}$  ?

**10.** Write each value in scientific notation.

**a.** 112 ten-thousandths

**b.** 0.00112

**c.** 112 billionths

**11.** Which is 0.00067 written in scientific notation?

- ☐ A.  $6.7 \times 10^{-4}$
- ☐ B.  $6.7 \times 10^{-3}$
- ☐ C.  $67 \times 10^{-5}$
- ☐ D.  $6.7 \times 10^4$

**12.** Select ALL of the expressions that are equal in value to 0.0034.

- ☐ A.  $3.4 \times 10^{-3}$
- ☐ B.  $34 \times 10^{-4}$
- ☐ C.  $0.34 \times 10^{-2}$
- ☐ D.  $340 \times 10^{-5}$
- ☐ E.  $3.4 \times 10^{-2}$
- ☐ F.  $3.4 \times 10^3$

**13.** A grain of pollen has a mass of about  $8 \times 10^{-9}$  gram. A bag of flour has a mass of about  $2 \times 10^3$  grams.

**a.** How many times larger is the mass of the bag of flour? Write the answer in scientific notation.

**b.** How many times smaller is the grain of pollen than the bag of flour?